

MASTER ARCHITECTURE SUMMARY

NextGen Mobile & Web Hybrid Application

Built for the FNB App Academy and South African Youth Wellness initiative under Indaba Men's Corner. Combines offline-first React SPA architectures with hardware-native Android runtime bridges via Capacitor.

1. CORE TOOLCHAIN & PREREQUISITES INVENTORY

Tool / SDK	Component / Package	Role in NextGen Project
Node.js & npm	Node v18+ / v20+	Build execution runtime and package manager
Vite & React	react, react-dom, vite	Declarative UI rendering, fast HMR, and bundle optimization
Capacitor Core	@capacitor/core, @capacitor/cli	Cross-platform native runtime bridge
Capacitor Android	@capacitor/android	Native Android platform container & Gradle runner
Device Sensors	@capacitor/geolocation	Live GPS coordinates for Family SOS distress beacon
Device Contacts	@capacitor-community/contacts	Secure local phonebook matching for safe friends chat
Storage Engine	@capacitor/preferences	Encrypted native key-value state persistence (POPIA)
Design System	lucide-react, Custom Glassmorphism	Ultra-gloss iconography, specular highlight UI tokens
Android Studio	Android SDK 34, JDK 17, Gradle	Emulator testing, signing, and AAB/APK release packaging

2. COMPLETE PROJECT BOOTSTRAPPING SEQUENCE

Step 1: Scaffold fresh React application using Vite

```
npm.cmd create vite@latest nextgen-app -- --template react
cd nextgen-app
```

Step 2: Install UI dependencies and Lucide icon library

```
npm.cmd install lucide-react
```

Step 3: Install Capacitor native core and CLI tools

```
npm.cmd install @capacitor/core
npm.cmd install -D @capacitor/cli
```

Step 4: Initialize Capacitor config (App Name & Package ID)

```
npx.cmd cap init "NextGen" "com.indabamenscorner.nextgen" --web-dir dist
```

Step 5: Install native hardware and storage plugins

```
npm.cmd install @capacitor/geolocation @capacitor-community/contacts @capacitor/preferences
npm.cmd install @capacitor/android
```

Step 6: Inject Android native platform directory

```
npx.cmd cap add android
```

3. DAILY DEVELOPMENT & LIVE HOT-RELOAD WORKFLOW

Run local development server with instant HMR:

```
cd C:\nextgen-app
npm.cmd run dev
```

Accessible locally at <http://localhost:5173/>

4. PRODUCTION BUILD & NATIVE SYNC SEQUENCE

Execute these two commands whenever changes are made to `src/App.jsx` or public assets:

1. Compile optimized web assets into `/dist` folder:

```
npm.cmd run build
```

2. Copy dist bundle and sync native plugin configurations to Android Studio:

```
npx.cmd cap sync android
```

5. ANDROID STUDIO LAUNCH & CLI RELEASE PACKAGING

Option A: Open the native project directly inside Android Studio GUI:

```
npx.cmd cap open android
```

Option B: Build production APK directly from terminal (No GUI required):

```
cd C:\nextgen-app\android
.\gradlew.bat assembleRelease
```

Option C: Build Google Play Store production bundle (.aab):

```
cd C:\nextgen-app\android
.\gradlew.bat bundleRelease
```

Inspect output directory in Windows Explorer:

```
explorer.exe app\build\outputs\apk\release
```

6. ANDROID PERMISSIONS & NATIVE MANIFEST RULES

Configured in `android/app/src/main/AndroidManifest.xml`:

- `android.permission.ACCESS_FINE_LOCATION` - High-precision GPS for Family SOS beacons
- `android.permission.ACCESS_COARSE_LOCATION` - Cellular tower positioning fallback
- `android.permission.READ_CONTACTS` - Local address book discovery for safe friends chat
- `android.permission.INTERNET` - WebView asset delivery and external social handoffs

7. GITHUB PAGES AUTOMATED LIVE WEB DEPLOYMENT

Build and push updates to the live web portal (<https://luckylah1.github.io/nextgen-app/>):

```
cd C:\nextgen-app
npm.cmd run build
git add .
git commit -m "feat: ultra-gloss UI upgrade, grade-adaptive companion, and social chat"
git push origin main
```

8. FULL SYSTEM RESET & CLEAN-BUILD PROCEDURES

If Gradle hangs, dependencies conflict, or Vite exhibits caching errors, run this exact sequence:

1. Purge Gradle cache and build artifacts:

```
cd C:\nextgen-app\android
.\gradlew.bat clean
cd ..
```

2. Reinstall Node modules cleanly:

```
rmdir /s /q node_modules
del package-lock.json
npm.cmd install
```

3. Re-bundle and re-sync native bridge:

```
npm.cmd run build
npx.cmd cap sync android
```

9. PRODUCTION RELEASE ARTIFACT LOCATIONS

Artifact Type	Filesystem Location / URL
Release APK	C:\nextgen-app\android\app\build\outputs\apk\release\app-release.apk
Production AAB	C:\nextgen-app\android\app\build\outputs\bundle\release\app-release.aab
Live Web App	https://luckylah1.github.io/nextgen-app/
Privacy Policy	https://luckylah1.github.io/nextgen-app/privacy.html
Pitch Deck PDF	C:\nextgen-app\NextGen_Pitch_Deck.pdf

10. KEY ARCHITECTURE CHECKPOINTS TO REMEMBER

- **POPIA Data Privacy:** User reflections, budget states, and dilemma votes must stay in @capacitor/preferences. Never transmit minor data to unauthenticated servers.
- **Social Handoff Deep-Links:** Uses WhatsApp (<https://wa.me/<phone>?text=...>), Instagram (<https://instagram.com/direct/inbox/>), and Messenger (<https://m.me/>).
- **Inner Companion Personas:** Controlled in COMPANION_GRADE_PERSONAS dictionary in src/App.jsx. Easily adjust copy for different grades or school campaigns.